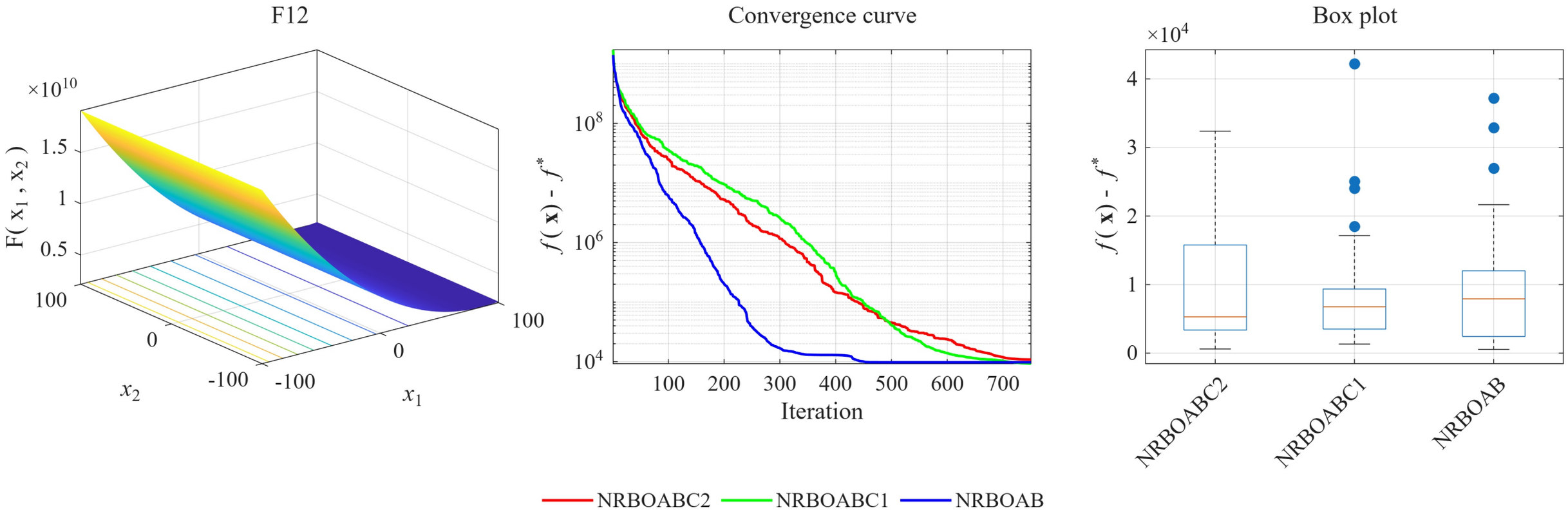

F12
×10^10
1.5
1
0.5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^8
10^6
10^4
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
4
3
2
1
0
f(x) - f*
NRBOABC2
NRBOABC1
NRBOAB
NRBOABC2
NRBOABC1
NRBOAB